This assignment will not be graded and is only for practice.

Level 1

1) Which of the following vectors can be written as the linear combination of vectors $(2, 3)$ and $(1, 2)$ in $\mathbb{R}^2$ with usual addition and scalar multiplication? ***1 point***

- [ ] $(1, 0)$
- [ ] $(0, 1)$
- [ ] $(0, 0)$
- [ ] Any vector in $\mathbb{R}^2$

No, the answer is incorrect.

Score: 0

Accepted Answers:

$(1, 0)$
$(0, 1)$
$(0, 0)$
Any vector in $\mathbb{R}^2$

2) Suppose the set $\{(3, 7, 4), v\}$ is a linearly dependent set in the vector space $\mathbb{R}^3$ with usual addition and scalar multiplication. Which of the following vectors are possible as a candidate for v? ***1 point***

- [ ] $(0, 0, 0)$
- [ ] $(5, \frac{35}{3}, \frac{20}{3})$
- [ ] $(1, 0, 0)$
- [ ] $(6, 14, -8)$

No, the answer is incorrect.

Score: 0

Accepted Answers:

$(0, 0, 0)$
$(5, \frac{35}{3}, \frac{20}{3})$

3) Suppose the set $\{(2,3,0),(0,1,0),v\}$ is a linearly dependent set in the vector space $\mathbb{R}^3$ with usual addition and scalar multiplication. Which of the following vectors are possible as a candidate for v? ***1 point***

- [ ] $(2,3,1)$
- [ ] $(1,2,0)$
- [ ] $(1,3,0)$
- [ ] $(0,1,1)$
- [ ] $(\pi,e,0)$

No, the answer is incorrect.
Score: 0

Accepted Answers:

$(1,2,0)$
$(1,3,0)$
$(\pi,e,0)$

4) If $(3,4)=\alpha(1,2)+\beta(1,3)$ in the vector space $\mathbb{R}^2$ with usual addition and scalar multiplication, then find the value of $\alpha+3\beta$.

No, the answer is incorrect.
Score: 0

Accepted Answers:

(Type: Numeric) -1

1 point

5) If (9, 3, 1) is a linear combination of the vectors $(1,1,1),(1,-1,1)$ and (x,y,z) as $2(1,1,1)+3(1,-1,1)+4(x,y,z)=(9,3,1)$, then find the value of $x+y+2z$.

No, the answer is incorrect.
Score: 0

Accepted Answers:

(Type: Numeric) 0

1 point

6) Let V be a vector space and v_1, v_2, v_3 and $v_4 \in V$. If v_1 is a linear combination of $v_i, i = 2, 3, 4$ $i.e.$ $v_1 = av_2 + bv_3 + cv_4$ then ***1 point***

- [ ] v_2 is a linear combination of $v_i, i = 1, 3, 4$.
- [ ] v_3 is a linear combination of $v_i, i = 1, 4$.
- [ ] v_2 is a linear combination of v_2.
- [ ] v_4 is a linear combination of v_4.

No, the answer is incorrect.

Score: 0

Accepted Answers:

v_2 is a linear combination of v_2.
v_4 is a linear combination of v_4.

7) Consider three vectors $v_1 = (1, 0, 0), v_2 = (1, 1, 0), v_3 = (1, 1, 1)$ in the vector space R^3, with usual addition and scalar multiplication. Which of the following sets is (are) true? ***1 point***

- [ ] $\{v_1, v_2, v_3\}$ is a linearly dependent set.
- [ ] $\{v_1 + v_2, v_1 + v_3, v_3\}$ is a linearly dependent set.
- [ ] $\{v_2 - v_1, v_1, v_2\}$ is a linearly dependent set.
- [ ] $\{v_1, v_2 - v_1, v_3 - v_2, v_3\}$ is a linearly dependent set.

No, the answer is incorrect.

Score: 0

Accepted Answers:

$\{v_2 - v_1, v_1, v_2\}$ is a linearly dependent set.
$\{v_1, v_2 - v_1, v_3 - v_2, v_3\}$ is a linearly dependent set.

Level - 2

8) Let S be a subset of $\mathbb{R}^3$ which is linearly dependent. Which of the following options are true? ***1 point***

- [ ] $S \cup \{(1,0,0)\}$ must be linearly dependent.
- [ ] $S \cup \{v\}$ must be linearly dependent for any $v \in \mathbb{R}^3$.
- [ ] $S \setminus \{v\}$ must be linearly dependent for any $v \in \mathbb{R}^3$.
- [ ] There may exist some $v \in S$, such that $S \setminus \{v\}$ is still linearly dependent.

No, the answer is incorrect.

Score: 0

Accepted Answers:

$S \cup \{(1,0,0)\}$ must be linearly dependent.
$S \cup \{v\}$ must be linearly dependent for any $v \in \mathbb{R}^3$.
There may exist some $v \in S$, such that $S \setminus \{v\}$ is still linearly dependent.

$M_{3\times3}(\mathrm{R})$ denotes the vector space consisting of real square matrices of order 3 with usual matrix addition and scalar multiplication. Consider the following matrices: $M_1 = \begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 0 \\ 1 & 1 & 1 \end{bmatrix}$, $M_2 = \begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 1 \\ 1 & 0 & 1 \end{bmatrix}$, $M_3 = \begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & -1 \\ 1 & 2 & 1 \end{bmatrix}$, $M_4 = \begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 2 \\ 1 & -1 & 1 \end{bmatrix}$

Use the above information to answer questions 9 and 10.

9) Which of the following options are correct? ***1 point***

- ☐ $M_1 - M_2 = 0$, where 0 denotes the zero matrix of order 3.
- ☐ $2M_1 + M_2 = M_3$
- ☐ $2M_1 - M_2 - M_3 = 0$, where 0 denotes the zero matrix of order 3.
- ☐ $2M_2 - M_1 - M_4 = 0$, where 0 denotes the zero matrix of order 3.
- ☐ The tuple (a, b, c) is unique for which $aM_1 + bM_3 + cM_4 = 0$ holds, where 0 denotes the zero matrix of order 3.
- ☐ The pair (a, b) is unique for which $aM_1 + bM_2 = 0$ holds, where 0 denotes the zero matrix of order 3.

No, the answer is incorrect.

Score: 0

Accepted Answers:

$2M_1 - M_2 - M_3 = 0$, where 0 denotes the zero matrix of order 3.
$2M_2 - M_1 - M_4 = 0$, where 0 denotes the zero matrix of order 3.
The pair (a, b) is unique for which $aM_1 + bM_2 = 0$ holds, where 0 denotes the zero matrix of order 3.

10) Which of the following options is(are) true? ***1 point***

- ☐ $\{M_1, M_2\}$ is a linearly dependent set.
- ☐ $\{M_1, M_2, M_3\}$ is a linearly dependent set.
- ☐ $\{M_1, M_2, M_4\}$ is a linearly dependent set.

- [ ] $\{M_1, M_3, M_4\}$ is a linearly dependent set.

No, the answer is incorrect.

Score: 0

Accepted Answers:

$\{M_1, M_2, M_3\}$ is a linearly dependent set.
$\{M_1, M_2, M_4\}$ is a linearly dependent set.
$\{M_1, M_3, M_4\}$ is a linearly dependent set.

Your score is: 0/10